

P24D8

Diagnostic information - DME (DFI) control unit

Tank vent pressure sensor: electrical fault

Diagnostic conditions

- Ignition on or engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Ground supply faulty
- ♦ Tank vent pressure sensor faulty
- For fault code P24D6:
 - ♦ Check valve between tank vent valve and intake manifold leaking or installed backward

P250C

Diagnostic information - DME (DFI) control unit

Oil level sensor

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to ground or open circuit in the lines
- ♦ Oil level sensor faulty

P250D

Diagnostic information - DME (DFI) control unit

Oil level sensor

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to ground or open circuit in the lines
- ♦ Oil level sensor faulty

P2531

Diagnostic information - DME (DFI) control unit

Power supply, terminal 15 or 87

Diagnostic conditions

- Ignition on
- or
- Engine running

Possible fault causes

- ♦ Vehicle electrical system supply voltage too high (e.g. jump-lead starting)
- ♦ Vehicle electrical system supply voltage too low
- ♦ Contact resistance, loose contact or corrosion on connectors or lines (DME (DFI) control unit)

Fault setting condition(s)

- Deviation in supply voltage at terminal 15 or 87:

- ♦ < 10 V (P0658/P2531)

or

- ♦ > 16 V (P0659/P2532)

at:

- ♦ DME (DFI) control unit - connector [B], contact 50 ⇒ Terminal 15
 - ♦ DME (DFI) control unit - connector [B], contact 3/5/6 ⇒ Terminal 87

P2532

Diagnostic information - DME (DFI) control unit

Power supply, terminal 15 or 87

Diagnostic conditions

- Ignition on
- or
- Engine running

Possible fault causes

- ♦ Vehicle electrical system supply voltage too high (e.g. jump-lead starting)
- ♦ Vehicle electrical system supply voltage too low
- ♦ Contact resistance, loose contact or corrosion on connectors or lines (DME (DFI) control unit)

Fault setting condition(s)

- Deviation in supply voltage at terminal 15 or 87:
 - ♦ < 10 V (P0658/P2531)or
 - ♦ > 16 V (P0659/P2532)
- at:
- ♦ DME (DFI) control unit - connector [B], contact 50 ⇒ Terminal 15
 - ♦ DME (DFI) control unit - connector [B], contact 3/5/6 ⇒ Terminal 87

P2539

Diagnostic information - DME (DFI) control unit

Fuel low-pressure sensor: electrical fault

Diagnostic conditions

- Ignition on
- or
- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Fuel low-pressure sensor faulty

P2541

Diagnostic information - DME (DFI) control unit

Fuel low-pressure sensor: electrical fault

Diagnostic conditions

- Ignition on
- or
- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Fuel low-pressure sensor faulty

P2545

Diagnostic information - DME (DFI) control unit

Torque intervention from transmission control unit implausible

Diagnostic conditions

- Drive with gear changes

Possible fault causes

- ♦ Shifting in PDK takes too long

Fault setting condition(s)

- The torque intervention by the PDK control unit takes too long; it is therefore implausible for the DME control unit.

P2610

Diagnostic information - DME (DFI) control unit

Switch-off time fault

Diagnostic conditions

- Engine running
- Then switch off ignition
- Immediately afterwards: ignition on > 10 minute(s)

Possible fault causes

- ♦ The power supply to the DME control unit was interrupted during control unit run-on (see fault code P0610 and U3001)
- ♦ DME control unit faulty

INFORMATION

The switch-off time is calculated in the DME control unit.

- ♦ The DME control unit receives time information from the instrument cluster at regular intervals of approx. 4 seconds when the ignition is on.
- ♦ These times are compared.

Diagnostic information - DME (DFI) control unit

Oxygen sensor upstream of catalytic converter, bank 1

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

i INFORMATION

♦ Do not interchange the oxygen sensor upstream of the catalytic converter with the sensor downstream of the catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Idling > 3 second(s)

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Oxygen sensor upstream of catalytic converter bank 1 faulty

Diagnostic information - DME (DFI) control unit

Oxygen sensor upstream of catalytic converter, bank 2

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

i INFORMATION

♦ Do not interchange the oxygen sensor upstream of the catalytic converter with the sensor downstream of the catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Idling > 3 second(s)

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Oxygen sensor upstream of catalytic converter bank 2 faulty

Diagnostic information - DME (DFI) control unit

Starter relay activation

Diagnostic conditions

- **PDK only:** Brake pressed + selector-lever position P or N
- **Manual transmission only:** Clutch pressed down fully
- Ignition on > 3 second(s)
- then start the engine

Possible fault causes

- ♦ Short circuit to B+
- ♦ Starter relay faulty

Fault effects

- ♦ Starter does not turn

Diagnostic information - DME (DFI) control unit

Starter relay activation

at:

- Starter turns during start attempt ► See **i** 1
- Starter does NOT turn during start attempt ► See **i** 2

i INFORMATION 1

Diagnostic conditions

- Ignition on > 3 second(s)
- then start the engine

Possible fault causes

- ♦ Short circuit to ground in starter relay activation (DME (DFI) control unit)
- ♦ Starter relay faulty

Fault effects

- **Manual transmission only:**
 - ♦ Engine start possible, with:
 - Clutch pedal not pressed
- **PDK only:**
 - ♦ Engine start possible, with:
 - Brake pedal not pressed
 - or
 - Selector-lever position R or D or M

i INFORMATION 2

Diagnostic conditions

- **PDK only:** Brake pressed + selector-lever position P or N
- **Manual transmission only:** Clutch pressed down fully
- Ignition on > 3 second(s)
- then start the engine

Possible fault causes

- ♦ Open circuit in starter relay activation (DME (DFI) control unit)
- ♦ Starter relay faulty

Fault effects

- ♦ Starter does NOT turn during start attempt

Diagnostic information - DME (DFI) control unit

Communication with drive CAN

Possible fault causes

- ◆ Short circuit in CAN-High line to ground
- ◆ Short circuit in CAN-High line to B+
- ◆ Short circuit in CAN-High line to interference signal
- ◆ Open circuit in CAN-High line

- ◆ Short circuit in CAN-Low line to ground
- ◆ Short circuit in CAN-Low line to B+
- ◆ Short circuit in CAN-Low line to interference signal
- ◆ Open circuit in CAN-Low line

- ◆ Faulty control unit in the designated drive CAN integrated system

Fault effects

- ◆ General vehicle functions are not available
- ◆ Existing substitute/emergency functions in the control units are activated
- ◆ Functions in the control units are not available or are available only with restrictions

Diagnostic information - valid for all control units

Communication with control unit

INFORMATION

Communication with the control unit designated in the fault text is/was not possible.

Possible fault causes

- ♦ The power supply for the designated control unit is not present or is present only to a limited extent.
- ♦ The CAN line CAN-High and/or CAN-Low is interrupted between the control unit connector of the designated control unit and the first splice behind the control unit.
- ♦ Control unit faulty

Fault effects

- ♦ Cross-vehicle functions are not available
- ♦ Existing substitute/emergency functions of the control units are active
- ♦ Functions in the control units are available only with restrictions
- ♦ Functions in the control units are not available

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Communication with control unit

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U0128

Diagnostic information - valid for all control units

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Fault effects

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U0159

Diagnostic information - valid for all control units

Communication with control unit

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Communication with control unit

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Diagnostic information - valid for all control units

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Fault effects

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U0300

Diagnostic information - DME (DFI) control unit

DME control unit programming

Diagnostic conditions

- Ignition on

Possible fault causes

- ◆ The DME programming version does not match the vehicle equipment.
- Program the **DME control unit** with the latest Porsche PIWIS Tester version.

Diagnostic information - DME (DFI) control unit

Coding incorrect

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Vehicle data incorrect
(See PIWIS Tester ► Additional menu [F7] ► Maintenance of vehicle data)
- ♦ Vehicle equipment wrong
- ♦ DME (DFI) control unit coded incorrectly - The data record does not match the vehicle.
- ♦ Specified control unit coded incorrectly

Diagnostic information - DME (DFI) control unit

Coding incorrect

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Vehicle data incorrect
(See PIWIS Tester ► Additional menu [F7] ► Maintenance of vehicle data)
- ♦ Vehicle equipment wrong
- ♦ DME (DFI) control unit coded incorrectly - The data record does not match the vehicle.
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Diagnostic information - valid for all control units

Communication with control unit

INFORMATION

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Possible fault causes

- ♦ The power supply for the designated control unit is not present or is present only to a limited extent.
- ♦ The CAN line CAN-High and/or CAN-Low is interrupted between the control unit connector of the designated control unit and the first splice behind the control unit.
- ♦ Control unit faulty

Fault effects

- ♦ Cross-vehicle functions are not available
- ♦ Existing substitute/emergency functions of the control units are active
- ♦ Functions in the control units are available only with restrictions
- ♦ Functions in the control units are not available

Diagnostic information - valid for all control units

Read out control unit fault memory content

INFORMATION

This fault code indicates one or more faults/notes in the fault memory of the designated control unit.

Existing faults/notes in the designated control unit must be processed first.

NOTE

Use the [F7] button to show/hide notes in the fault memory.

Possible fault causes

- ◆ Fault in the designated control unit
- ◆ Note in the designated control unit

Fault effects

- ◆ Functions in the control unit are available only with restrictions
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Diagnostic information - valid for all control units

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